

AI-Assisted Drafting of ISO 9001 / AS9100 Work Instructions Using Microsoft Foundry

TechCoaches Information Systems Design (fictional) · Aerospace & Defense Documentation

PREPARED BY	STANDARDS	FORMAT	STATUS
Nathan Smith	ISO 9001:2015; AS9100D	DITA 1.3 (concept, task, reference, glosssgroup)	Hypothetical / Demonstration

DITAMAP STRUCTURE · WI-QA-8.5-014.DITAMAP

- Overview (concept)
- **Concepts**
 - Data Handling and Security Boundaries (concept)
- **Reference**
 - Roles and Responsibilities (reference)
- **Tasks**
 - Configure a Foundry Project for WI Drafting (task)
 - Draft a Work Instruction Using the Foundry Pipeline (task)
- **Records**
 - Required Records and Retention (reference)
- **Glossary**
 - Glossary of Terms (glosssgroup)

AI-Assisted Work Instruction Drafting—Overview

Establishes the purpose, scope, and quality management system basis for using Microsoft Foundry as a drafting aid for Work Instructions, and defines the boundary between AI-assisted drafting and the human verification required for release.

Purpose

This Work Instruction defines the approved pipeline for using Microsoft Foundry (Azure AI Foundry) to accelerate the drafting of Work Instructions, forms, and guidance documents that must conform to ISO 9001:2015 and AS9100D, while keeping every AI-assisted step inside the review, verification, and record-keeping controls the quality management system already requires for documented information.

Scope

Applies to Technical Authors, SME reviewers, and Document Control representatives who create or revise Work Instructions, forms, and guidance documents using the Foundry Work Instruction Drafting Project. Covers text drafting only — not engineering design outputs, CAD/model-based definition data, or content classified above Company Proprietary.

Quality System Basis

CLAUSE	BASIS
ISO 9001:2015 §7.5	Documented Information — AI-assisted drafts are subject to the same identification, review, approval, and control as any other documented information.
ISO 9001:2015 §8.5.1	Control of Production and Service Provision — documentation is itself a service output; its production process, including any AI tooling used within it, must be controlled.
AS9100D §8.5.1.1	Control of Production Equipment, Tools, and Software Programs — the Foundry drafting project is software used within a service provision process and must be validated and periodically revalidated as fit for its intended purpose.

Governing principle: Microsoft Foundry may accelerate a draft. It may never be the sole authority for technical accuracy, safety content, or conformance. Every AI-assisted draft is verified by a human SME and reviewed by Document Control before release.

Data Handling and Security Boundaries for Microsoft Foundry Use

Defines which categories of source content may be entered into the Foundry Work Instruction Drafting Project, and the controls that must be in place first.

Data Classification Tiers

Permitted and Prohibited Source Content

TIER	EXAMPLES	FOUNDRY USE
Public / Released Unclassified	Approved, released Work Instructions; public-facing operator guides	Permitted
Company Proprietary	Internal process descriptions; unreleased drafts under change control	Permitted within the isolated project only
Export-Controlled (ITAR/EAR)	Technical data subject to ITAR/EAR	Prohibited unless authorized in writing by the Empowered Official / Export Control Officer
CUI / Classified	Controlled Unclassified Information; classified technical data	Prohibited — never entered under this Work Instruction

Network and Identity Controls

- Project provisioned behind a private endpoint; public network access disabled.
- Authentication via Microsoft Entra ID managed identities; no shared API keys.
- Content Safety filtering remains enabled on every deployment.
- Source-data connections limited to the approved document management/PLM connector; no open web browsing tools enabled.

If you are unsure: treat the document as Export-Controlled until the program lead confirms otherwise. When in doubt, do not paste it into Foundry.

Roles and Responsibilities

Defines who may request, configure, use, review, and approve output from the Foundry Work Instruction Drafting Project.

ROLE	RESPONSIBILITY
Technical Author	Classifies source content, submits it to the approved drafting flow, reviews raw AI output for fabricated content, incorporates SME and Document Control feedback.
SME Reviewer	Verifies every technical value, step, and condition against released engineering data; signs the SME Technical Verification Record.
Document Control Representative	Confirms numbering, revision, and distribution requirements; issues the controlled release.
AI Systems Administrator (Foundry Project Owner)	Provisions and maintains the hub/project, maintains the Approved Model List, configures Content Safety and network controls, performs periodic revalidation.
Export Control Officer / Empowered Official	Authorizes, in writing, any exception allowing export-controlled content into a specific deployment. No exception is implied by this instruction.

Configure a Microsoft Foundry Project for Work Instruction Drafting

How the AI Systems Administrator provisions and validates a Foundry project before it may be used to draft Work Instructions.

Satisfies **AS9100D §8.5.1.1**: a Foundry project used in a service provision process is production software and must be validated as fit for its intended purpose before use, and periodically revalidated afterward. Performed by: AI Systems Administrator.

1. Confirm authorization to provision the project.
Requires a signed request citing WI-QA-8.5-014 and the program(s) supported. Do not provision on an informal or verbal request.
2. Create or select a Foundry hub meeting the network requirements (private endpoint, public access disabled).
3. Create the Work Instruction Drafting project within that hub, named WI-DRAFTING-[program code].
4. Select an approved model from the model catalog.
Only models on the current Approved Model List. Confirm rather than assume.
5. Build the drafting flow using the approved prompt template, with no open web browsing tool attached.
6. Enable Content Safety filtering and confirm the threshold matches the standard configuration.
Do not disable filtering to work around a blocked response. Escalate for a documented, justified adjustment instead.
7. Restrict data connections to the approved document management/PLM connector only.
8. Validate the configuration against the Foundry Project Validation Checklist and record model version, network configuration, and Content Safety settings in the AI Drafting Log.
9. Release the project for use and notify authorized Technical Authors.

Result: A validated Foundry project, with its configuration on record, ready for Technical Author use.

Draft a Work Instruction Using the Foundry Drafting Pipeline

The end-to-end steps a Technical Author follows to produce a conformant Work Instruction with Foundry assistance, from source content through controlled release.

Prerequisites

- A validated Foundry Work Instruction Drafting project exists.
- Source content has been classified and is Public/Released or Company Proprietary.
- A change request or new-document authorization exists.

1. Gather released source content (ECO, drawings, prior revision).
Do not draft from memory or an SME's verbal description alone; capture the input in writing first.
2. Open the drafting flow and enter document metadata (WI number, product line, process step).
3. Submit the source content using the approved prompt template, without modifying its constraints.
Example: state the target standard (e.g., STE), instruct the model to flag rather than resolve ambiguity, and forbid adding anything not present in the source.
4. Review the raw AI-generated draft before forwarding it.

IF THE DRAFT...	THEN...
Flags an ambiguity in the source	Resolve it with the requesting SME first. Do not guess.
Contains a value or step not in the source	Remove or correct it. Do not forward unverified, AI-introduced content.
Matches the source faithfully	Proceed to the format and compliance check.

5. Run the draft against the format and compliance checklist (hazard warnings, PPE, imperative-mood steps, verification points).
6. Route to the SME Reviewer for technical verification and sign-off.
7. Route the SME-verified draft to Document Control for conformance review and release.
8. Log the AI-drafting record: model/version, prompt template version, Technical Author, and both reviewers' sign-offs.
9. Release the Work Instruction into the document management system.

Result: A released, controlled Work Instruction, with a complete record of the AI assistance used and the human verification that preceded release.

Required Records and Retention

The records this Work Instruction requires, the clause each demonstrates conformance to, and the minimum retention period.

RECORD	DESCRIPTION	DEMONSTRATES CONFORMANCE TO	RETENTION
AI Drafting Log	Project, model/version, prompt template version, Technical Author, date	AS9100D §8.5.1.1	Life + 7 yrs
Foundry Project Validation Record	Evidence of validation before first use and on material change	AS9100D §8.5.1.1	Superseded + 3 yrs
SME Technical Verification Record	Signed confirmation of technical checks against released data	ISO 9001 §7.5; §8.5.1	Life + 7 yrs
Document Control Review & Release Record	Approval evidence for numbering, revision, distribution	ISO 9001 §7.5.3	Life + 7 yrs
Work Instruction Revision History	Change log identifying what changed and why	ISO 9001 §7.5.2	Life + 7 yrs

Glossary of Terms

Foundry project — A workspace within a Microsoft Foundry hub where a model, drafting flow, and connections are configured for a specific purpose.

Content Safety — The Foundry filtering service that screens prompts and generated output for harmful or policy-violating content.

Fabricated content (hallucination) — A value, step, or statement produced by an AI model that is not supported by the source content provided to it.

Subject Matter Expert (SME) — A qualified engineer, technician, or operator who verifies technical accuracy against released engineering data.

Controlled Unclassified Information (CUI) — Government unclassified information requiring safeguarding or dissemination controls; never entered into Foundry under this instruction.

Model catalog — The Foundry marketplace of available language models from which an approved model is selected for a project.

Source DITA files (the ditamap and all seven topics) accompany this rendered preview. This PDF exists for convenient reading; the DITA XML is the authoritative writing sample.